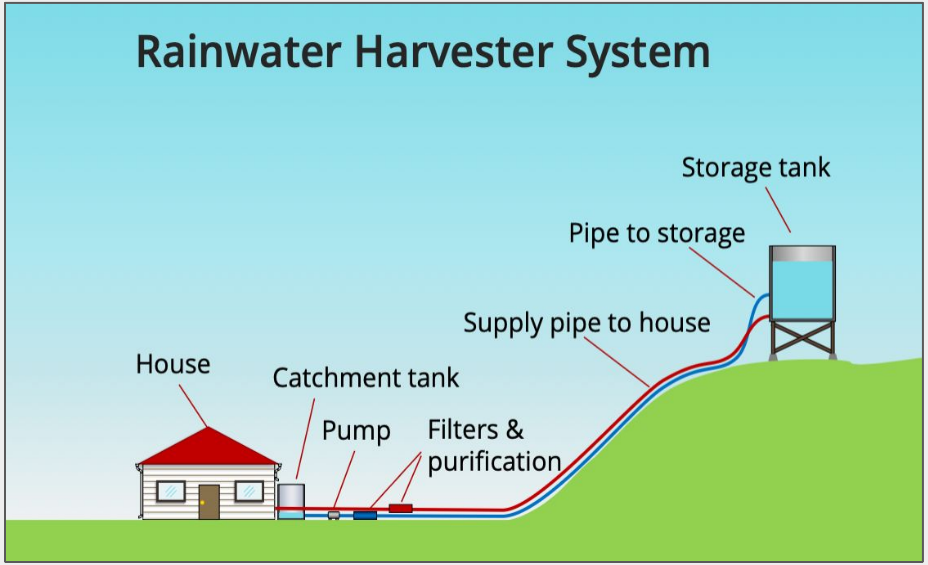
Rainwater Harvester System

Team B5

Context



- Van Anda, BC, needs a dependable way to provide potable water to households.

- Our goal: design a rainwater harvesting system that can collect, treat, store, and distribute potable water.

- System must be self-powered, meet daily water needs of a two-resident household, and be reliable over a 5-year period.

Overview

Our Design Solution

Decision Making Process

- *Stakeholder Consultation*
- *Virtual Simulation*

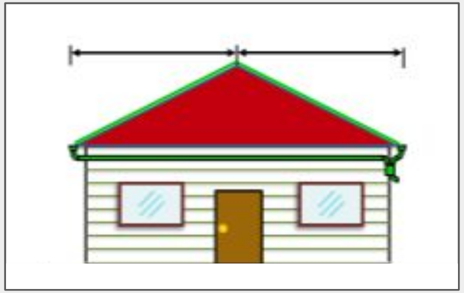
Calculations & Assumptions

- *Weather, Energy, Cost*

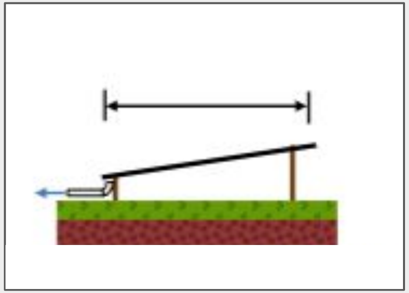
Stakeholder Satisfaction

Conclusion

Final Recommendation: Catchment & Pump



Whole Roof
100m^2



Additional Catchment
10m^2

Catchment Tank
2500L

Pump A
900 running hours

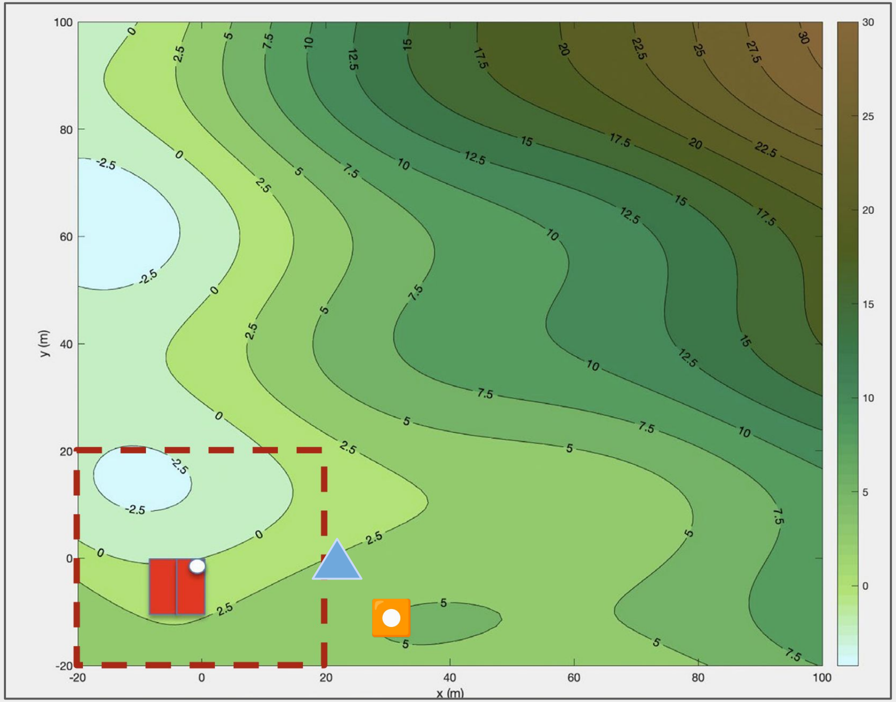
Final Recommendation: Storage

Storage Tank: $50m^3$

Storage Tank Location:

- 29m from House
- 5m Elevation

No additional storage tank tower



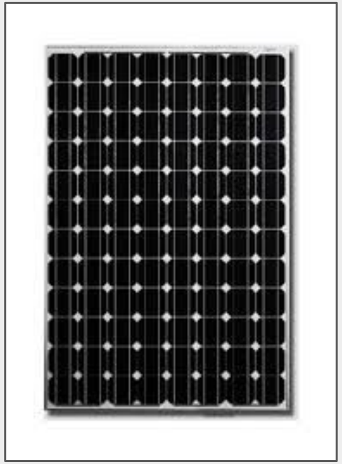
Storage tank location *Additional catchment location*

Final Recommendation: Water Treatment & Power

Disinfection
Ozone Generator & Injection System

UV System
36W

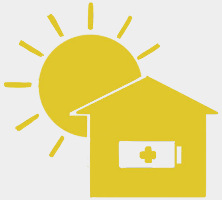
Additional Filtration
1µm, 5µm, 200µm
Placed on line from catchment to storage



Solar Power System
HES-260, 9 Panels
7 additional 12V Batteries

Overall Strategy

Primary Goal: Maximize Net Stakeholder Satisfaction



100% Off-Grid Independence

No reliance on external fuel supply chains

Health & Quality First

Superior water taste & zero chemical risk

Balanced Sustainability

Optimizing capital cost vs. GHG emissions

Decision Making Process

1) Stakeholder Consultation

- **Action:** Translated stakeholder interviews into measurable metrics.
- **Focus:** Prioritized stakeholder satisfaction targeting health, taste, and off-grid reliability.

2) Prototyping & Simulations

- **Action:** Utilized spreadsheet models to quantify system parameters.
- **Focus:** Calculated pump curves, friction losses, energy requirements, costs, and satisfaction values.

3) Develop & Refine Solution

- **Action:** Iterative adjustments to optimize parameters and satisfaction.
- **Focus:** Balanced a reliable system with a reasonable cost, sustainable parameters, and stakeholder needs.

4) Testing & Implementation

- **Action:** Virtual testing using 5-year simulation.
- **Focus:** Verified the design successfully sustains the 5-year simulation period with optimal satisfaction values.

Stakeholder Consultation

Primary Stakeholders: Van Anda Community Residents

Secondary & Tertiary Stakeholders: Community officials, BC government, component manufacturers and suppliers

Expressed Needs:

- Good water quality and taste
- Affordability
- 24/7 access to potable water
- Independent energy source
- Minimal GHG emissions
- Access to maintenance when needed
- Safe transportation and handling of hazardous materials

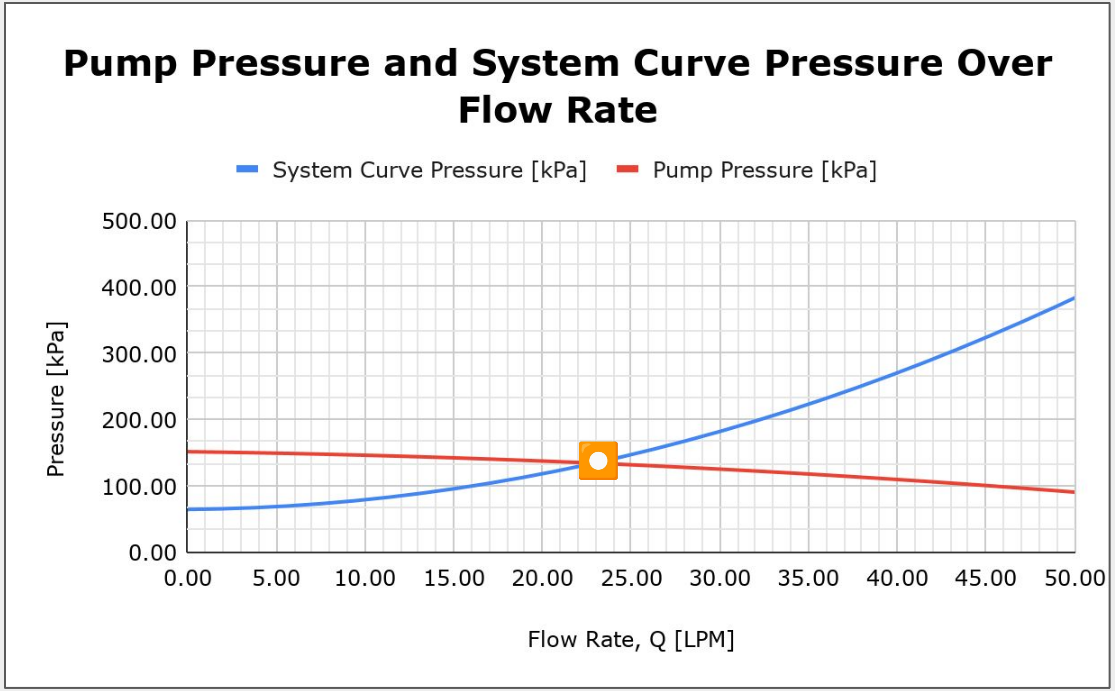
Alternative Configurations

Component Category	Rejected Alternative	Final Selection
1. Disinfection	Chlorine	Ozone + UV
2. Power Source	Diesel Generator	Solar (HES-260)
3. Tank Elevation	6m Storage Tower on 5m Hill	Ground Tank (on 5m Hill)
4. Catchment Area	Standard Roof Only	Roof + Additional
5. Pump Selection	Pump C (High Power)	Pump A

Alternative Configurations

Component Category	Rejected Alternative	Final Selection	Justification for Our Final Selection
1. Disinfection	Chlorine	Ozone + UV	Chlorine failed stakeholder need for "good water taste" & ozone improves risk satisfaction.
2. Power Source	Diesel Generator	Solar (HES-260)	Avoiding diesel handling mitigates risk and solar energy removes reliance on external fuel logistics in a remote area.
3. Tank Elevation	6m Storage Tower on 5m Hill	Ground Tank (on 5m Hill)	Lower on demand flow rate from ground tank is compensated by improved capital cost and reduced required pump pressure.
4. Catchment Area	Standard Roof Only	Roof + Additional	Additional catchment area allows more water to be collected in minimal rain conditions, avoiding dry times in a 5-year period.
5. Pump Selection	Pump C (High Power)	Pump A	Pump A has the highest efficiency at our pump flow rate, meaning it requires less solar energy, and has a low initial cost.

Pump Curve



System pressure accounts for elevation change and friction losses; increasing pressure.

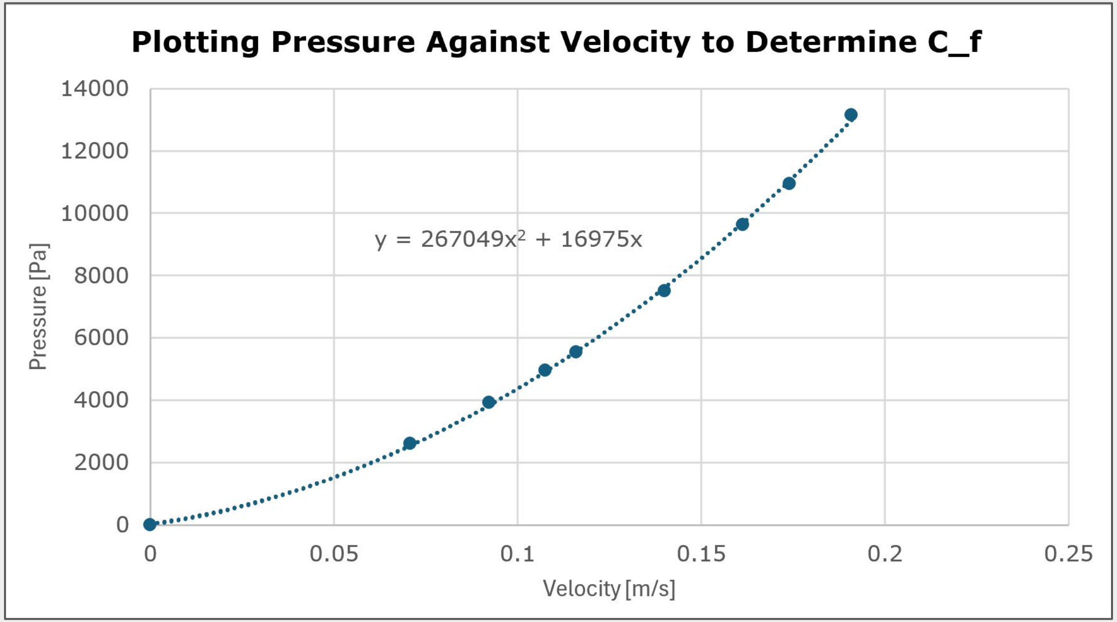
To achieve steady water flow:

Operating Pressure: 132 kPa

Max Flow Rate: 23 LPM

Filter Coefficient Analysis

Each filter introduces friction, which increases the amount of pressure the pump must exert to pump the water to the storage tank.



$C_f = 16975$ Pa/(m/s)

Loss from 1µm filter:
$0.167 * C_f$

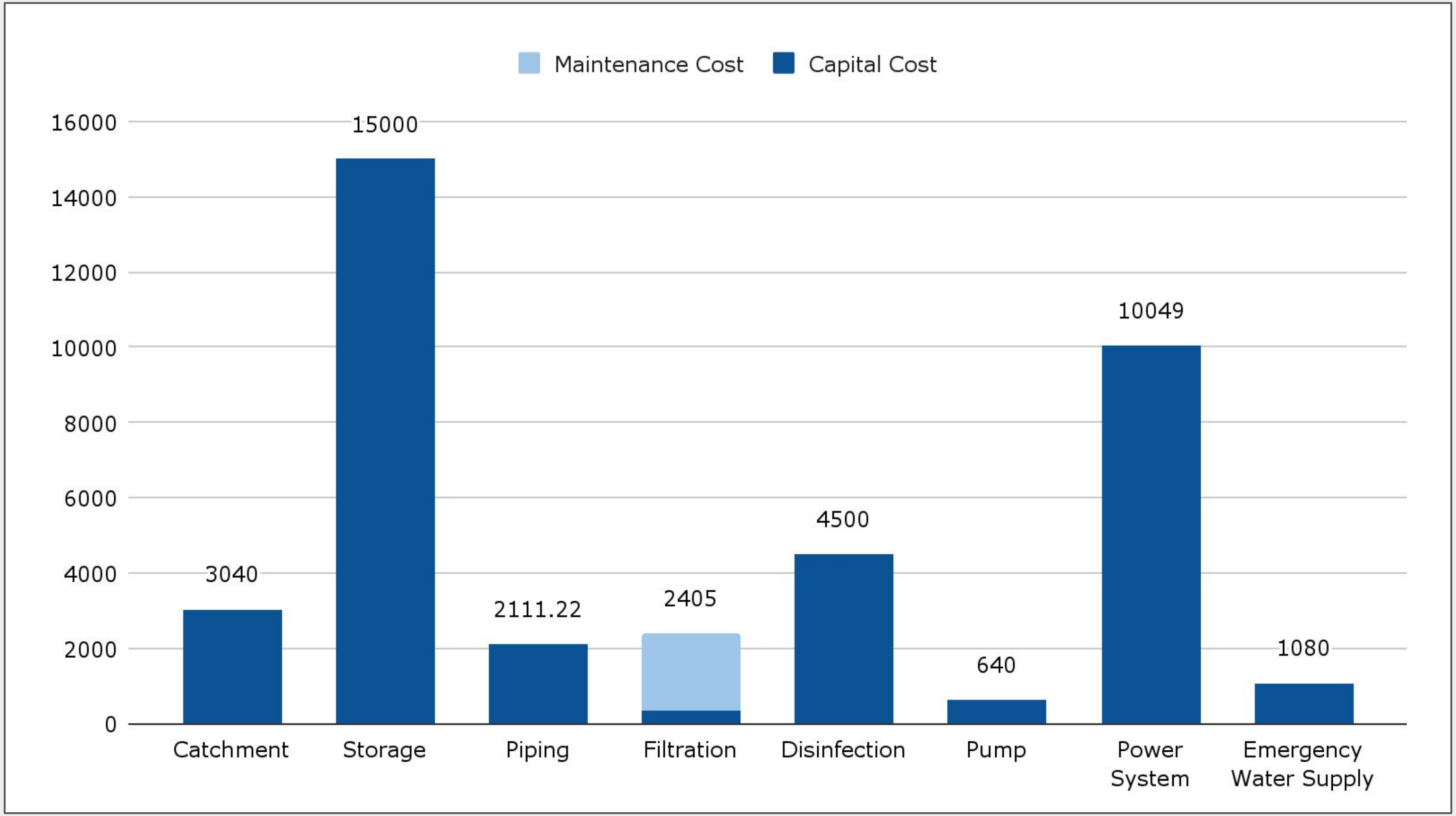
Loss from 5µm filter:
$0.0833 * C_f$

Loss from 5µm filter:
$0.00417 * C_f$

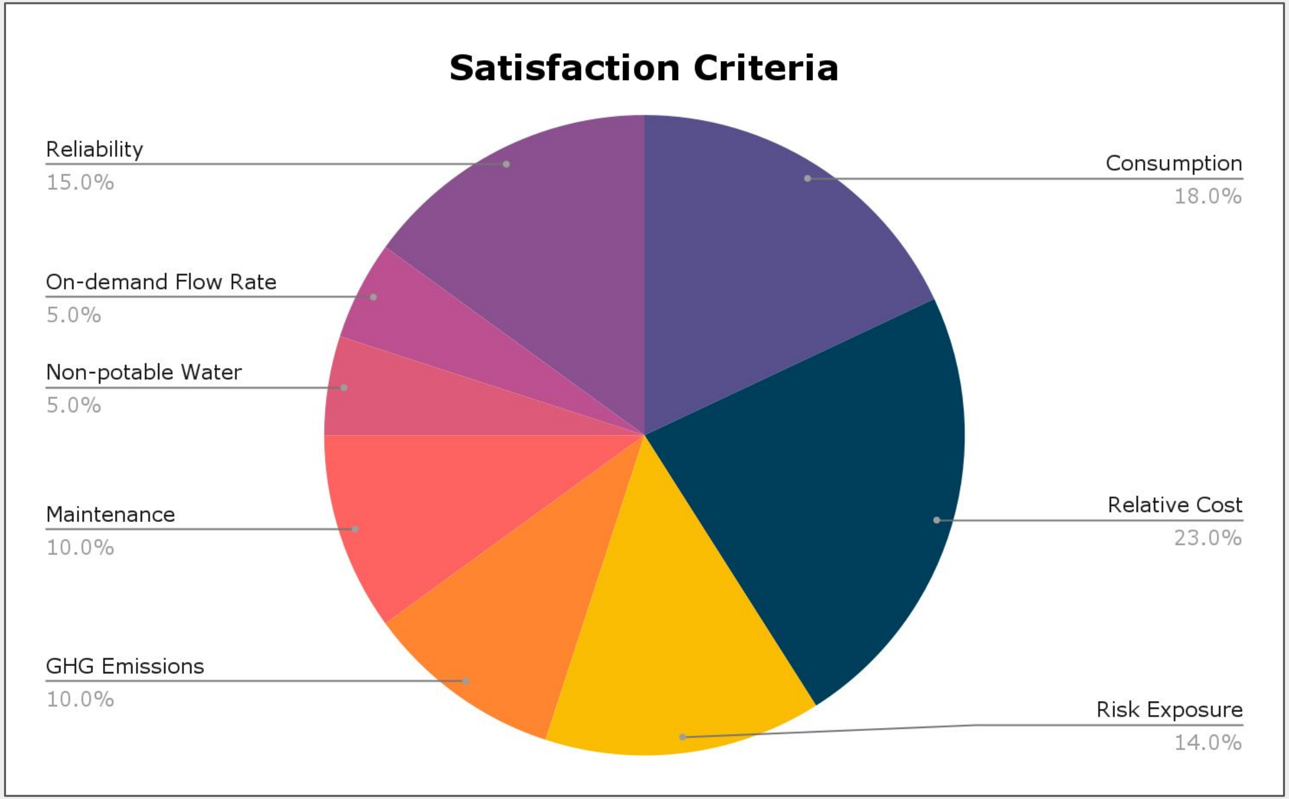
Acknowledgement: data set was provided and approved by APSC 101 Teaching Team

Cost Analysis

5 year total: $38,825.22

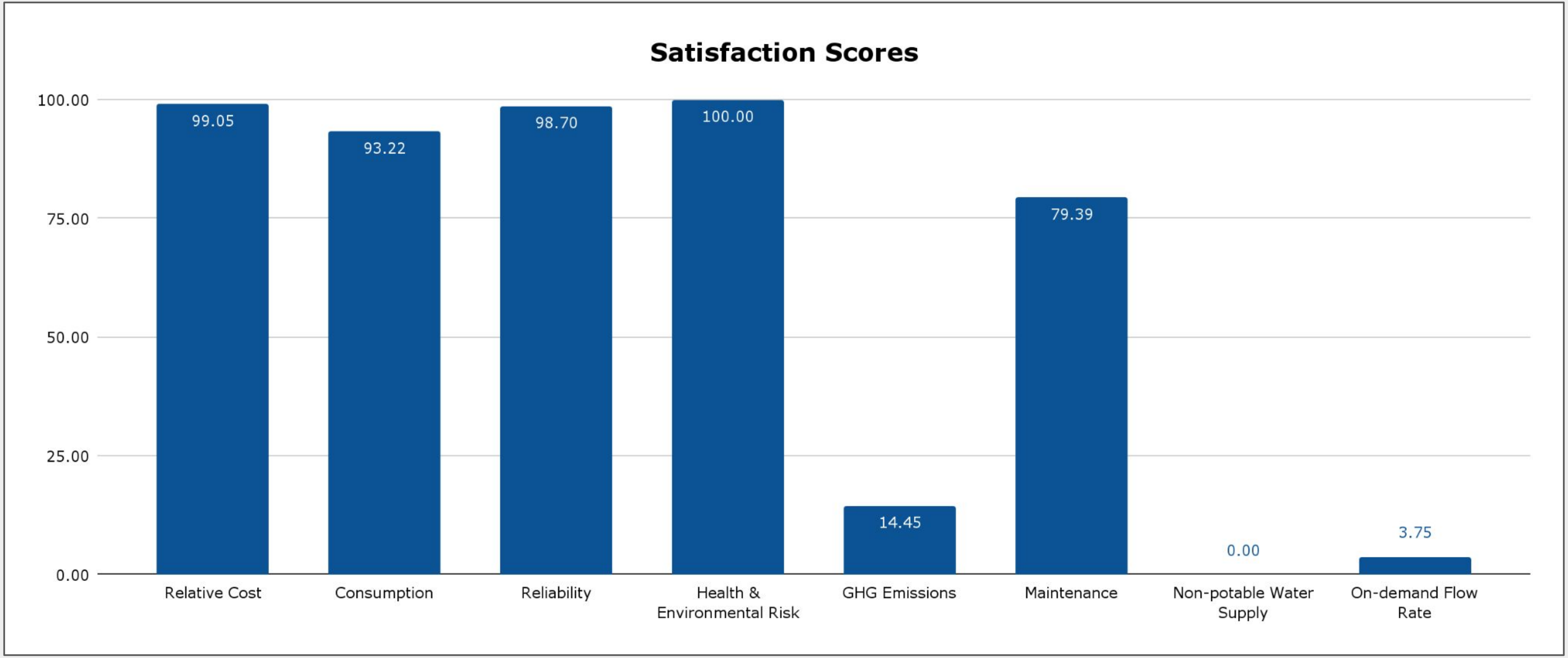


Based on the context and priorities of residents in the Van Anda community, satisfaction criteria were weighted as follows:



Criterion Satisfaction Scores

Overall Satisfaction: 77.94%



Conclusion

Reliable & Sustainable System

Set to provide safe drinking water for 5+ years whilst emitting minimal greenhouse gas emissions.

Balanced Cost & Performance

Our solution costs 36.22% of the relative cost to operate now, and provides on demand water when desired.

Safe, Potable Water Supply

Our design has a thorough filtration and disinfection process, without compromising water taste.

Stakeholder-centered Solution

This solution was designed to meet resident needs, and produces 77.94% stakeholder satisfaction.

Thank You

Presented by B5

There was no use of GenAI for any purposes in this presentation.